Unit 3: Signals and Systems

1. What do you mean by power type and energy type signals? Check whether x(t) = Acosω0t is power type or energy type signal. [4+4] 2

2. Determine if the following system described by y(t) = cos[x(t+2)] is causal, linear, time variant and stable. [7]

3. What is unit step function and impulse function? [4] 2011 [F], 2012 [S] ,2013 [F], 2016 [S] ,2017 [F], 2018 [F] ,2019 [F], 2019 [S], 2023 [F]

4. What do you mean by time invariant system? Check whether the function $\mathbf{Y[n] = n^2\ x[n\text{-}x(n\text{-}1)]}$ is time variant or invariant. [5]

5. Write basic properties of linearity, causality, stability and time invariance.

6. Differentiate between energy and power signal with example.

7. Define system. Is the system y[n] = n .x[n] is

i) linear

ii) time variant?

iii) causal or not.

8. Write short notes on:

- Unit step function & Impulse function.
- Power versus Energy signal
- continuous and discrete time signal.
- Deterministic vs Random Signal.

9. Determine if the system described by y(t) = cos[x(t+5)] is causal, linear, time variant and stable.

10. Is system described by characteristic equation: y(t) = A x(t) + B, determine whether the system [7]

i) linear or non-linear

ii) causal or non-causal

iii) stable or unstable

iv) static or dynamic.

11. Define LTI systems. How will you determine whether a signal is power signal or energy signal?

12. Justify whether $\mathbf{x(t)} = e^{-a|t|}$ for a > 0 is energy signal or power signal.

13. How would you say that a system is BIBO stable? Comment about the stability for the following system

$$\mathbf{y(n) = 2x(n+1) + [x(n-1)]^2}$$

14. How do you classify the signal? Explain why a sinusoidal wave is periodic and deterministic signal.

15. Check whether the given system is linear, causal and time invariant or not. [7]

i) y(t) = log x(t)

ii) y(t) = t* x(t)

iii) $\mathbf{y(t) = t^2 * x(t)}$

16. Comment about the linearity, stability, time-invariance and causality for the following system: [8] y(t) = t2x(t)

17. Find the energy and power of signal $\mathbf{x(t)} = e^{-4t}\mathbf{u(t)}$ and justify whether it is energy signal or power signal. [8]

18. Determine whether the signal x(t) = A[u(t+a)-u(t-a)] for a > 0 is a power signal or energy signal.

19. For a given signal x(t) = cosωt, find the magnitude and phase of spectrum.

20. Explain with examples, how periodic & non-periodic signals are distinguished. Justify whether ramp signal is energy signal or power signal.

21. Check whether x(t) = Acosωt, is power or energy type signal. [7] 2021 [S] [7] 2021[F] [7] 2023 [S]

22. Define Signal.Justify whether unit step signal is energy signal or power signal.

23. Define sinusoidal signal, Signum signal ,unit step signal, impulse and ramp signal graphilcally with their mathematical relation.